

Exercise Sheet 4

Centrality and Structural Roles

5942UE Network Science

Prof. Dr. Michael Granitzer Dr. Kanishika Ghosh Dastidar Dr. Jelena Mitrovic

Module A — Centrality Measures

Exercise 4.A.1: Centrality by Hand

Consider the graph with nodes A, B, C, D, E, F and edges $A-B, A-C, B-C, B-D, D-E, D-F, E-F$.

- 1. Compute **degree centrality** $C_d(v) = \deg(v)/(|V| - 1)$ for all nodes.
- 2. Identify the shortest paths between all node pairs. Which node lies on the most shortest paths between other nodes? (Compute betweenness centrality by inspection.)
- 3. Which node has the highest degree centrality? Which has the highest betweenness centrality? Are they the same node?
- 4. Interpret: what structural role does the node with the highest betweenness play in this network?

Solution:

1. Degree centrality ($|V| - 1 = 5$):

Node	Degree	C_d
A	2	2/5



Node	Degree	C_d
B	3	3/5
C	2	2/5
D	3	3/5
E	2	2/5
F	2	2/5

2. Betweenness — by inspection:

The graph has two triangle-like clusters: A, B, C on the left and D, E, F on the right, joined by edge $B-D$. Every shortest path from any node in A, C to any node in D, E, F must pass through B then D . Both B and D have very high betweenness; D slightly higher because it is the unique gateway to E, F .

3. Highest degree centrality: B and D are tied at $3/5$. **Highest betweenness:** D (and B close behind), but D edges to E and F which are a "terminal" cluster, making D the unique gateway.

Exercise 4.A.2: When Centrality Measures Disagree

Using `nx.karate_club_graph()`:

1. Compute degree, betweenness, closeness, and eigenvector centrality for all nodes.
2. Find the top-3 nodes for each measure. Do the top lists overlap?
3. Plot four subplots, each showing the network with node sizes proportional to one centrality measure.
4. Find a node that ranks **very differently** across measures. What does this reveal about its structural position?

Solution:

```
import networkx as nx
import matplotlib.pyplot as plt

G = nx.karate_club_graph()

# Centrality dashboard
centralities = {
    "Degree": nx.degree_centrality(G),
    "Betweenness": nx.betweenness_centrality(G),
    "Closeness": nx.closeness_centrality(G),
    "Eigenvector": nx.eigenvector_centrality(G),
}

# Top-3 per measure
for name, c in centralities.items():
```



```
top3 = sorted(c, key=c.get, reverse=True)[:3]  
print(f"{name}: {top3}")
```

Interpretation:

- **Leading hubs:** Nodes 0 and 33 lead in almost every measure — they are the faction leaders with high degree and influence.
- **Structural disagreement:** Node 32 often shows high betweenness relative to its degree because it acts as a bridge to node 33's faction. High eigenvector centrality in dense cores often pairs with low betweenness due to redundant paths.

Module B — Structural Roles

Exercise 4.B.1: Embedded vs. Broker

Consider a graph with two triangles 1, 2, 3 and 4, 5, 6 connected by edge 3–4, plus node 7 connected only to nodes 3 and 4.

1. Is node 1 **embedded** or a **broker**? What evidence supports your answer?
2. Is node 7 **embedded** or a **broker**? What is its clustering coefficient C_7 ?
3. If edge 3–4 is removed, what happens to node 7's structural role?
4. Burt (1992) argues brokers gain "information advantage." What unique information could node 7 access that nodes in 1, 2, 3 cannot?

Solution:

1. Node 1 — embedded: Node 1 is surrounded by the closed triangle 1, 2, 3 where edges 1–2, 1–3, 2–3 all exist. Its neighbours are mutually connected. Clustering coefficient $C_1 = 1$. Node 1 cannot reach 4, 5, 6 without going through 3 (and then 4), so it has no direct brokerage opportunity.

2. Node 7 — broker: Node 7 connects to 3 and 4, which are the two bridge points between the  triangles. Are 3 and 4 connected? Yes — edge 3–4 exists. So $C_7 = 1$ (the one possible edge between

7's two neighbours exists). Despite $C_7 = 1$, node 7 structurally straddles two groups: it can hear information from both the 1, 2, 3 cluster (via 3) and the 4, 5, 6 cluster (via 4).

3. If 3-4 is removed: Node 7 would then connect to 3 (in cluster 1, 2, 3) and 4 (in cluster 4, 5, 6) with no edge between its neighbours $\rightarrow C_7 = 0$. Now node 7 becomes the **only** bridge between the two components. Its brokerage role becomes critical.

4. Information advantage: Node 7 can learn about jobs, ideas, and events in the 4, 5, 6 cluster that are invisible to 1, 2, 3 (and vice versa), because information circulates within dense clusters but rarely crosses to the other side without a broker. This is Burt's "structural hole" argument: the gap between the two triangles is the structural hole, and 7 bridges it.

Exercise 4.B.2: Betweenness as a Proxy for Brokerage

Using the two-triangle + node 7 graph from Exercise 4.B.1:

1. Build the graph in NetworkX. Compute betweenness centrality for all nodes.
2. Verify that node 7 has high betweenness despite $C_7 = 1$.
3. Explain the apparent contradiction: high clustering coefficient yet high betweenness.
4. On the karate club graph: find the 3 nodes with the highest betweenness but **lowest** clustering coefficient. What does this combination mean structurally?

Solution:

```
import networkx as nx

G = nx.Graph()
G.add_edges_from([(1,2),(2,3),(1,3), (4,5),(5,6),(4,6), (3,4), (7,3),(7,4)])

bc = nx.betweenness_centrality(G, normalized=True)
cc = nx.clustering(G)

# High betweenness despite high clustering coefficient
# Score = betweenness / (clustering + 1e-6) - ranks bridge-brokers
K = nx.karate_club_graph()
bc_k = nx.betweenness_centrality(K)
cc_k = nx.clustering(K)
scores = {n: bc_k[n] / (cc_k[n] + 1e-6) for n in K.nodes()}
```



```
top3 = sorted(scores, key=scores.get, reverse=True)[:3]
print("Bridge-brokers:", top3)
```

Interpretation:

- **Metric contrast:** Clustering coefficient is a local density measure (neighbours connected?), while betweenness is a global routing measure (on shortest paths?).
- **Brokerage with high C:** Node 7 has $C_7 = 1$ because its two neighbours are tied, but it still has high betweenness because it provides an alternative path between two large clusters.
- **Pure bridges:** High betweenness with low clustering (like node 32 in the karate club) indicates a pure bridge connecting otherwise separate groups.